

RAG Framework Comparison Report

RAGking v2.1 - Multi-Framework Analysis

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Frameworks compared: 2 — n8n, lan

Framework Summary - Metric Definitions

Formulas Used

$$\text{RAGking Score} = \left(w_C \cdot \hat{C} + w_E \cdot \hat{E} + w_L \cdot \hat{L} + w_M \cdot \hat{M} + w_F \cdot \hat{F} \right) \times 100$$

$$\text{Cost}_{\text{query}} = \frac{T_{\text{in}} \cdot P_{\text{in}} + T_{\text{out}} \cdot P_{\text{out}}}{10^6}$$

$$\text{Economic Efficiency} = \frac{\text{Correctness}}{\text{Cost}_{\text{query}}}$$

$$\text{Token Efficiency} = \frac{T_{\text{in}} + T_{\text{out}}}{\max(\text{Correctness}, 1)}$$

$$\text{Failure Rate} = \frac{N_{\text{fail}}}{N_{\text{total}}} \times 100$$

Metric	Description	Unit
RAGking Score	Composite ranking score combining all performance dimensions	[0–100]
Correctness	LLM-as-a-Judge score measuring answer quality	[1–5]
Latency	Average response time from query to answer	[ms]
Similarity	Cosine similarity between generated and expected answers	[0–1]
Failure Rate	Percentage of invalid or empty responses	[%]
Token Efficiency	Total tokens consumed per correctness point (↓ better)	[tok/pt]
Std Dev	Standard deviation of RAGking Score (↓ better)	[score]
Prompt Tokens	Average number of input tokens per query	[tokens]
Completion Tokens	Average number of output tokens per response	[tokens]
Total Tokens	Sum of prompt and completion tokens	[tokens]
Deploy Time	Estimated hours to setup and deploy	[hours]

Metric	Description	Unit
Difficulty	Implementation complexity rating	[1–5]
Maintenance	Weekly maintenance effort required	[h/week]

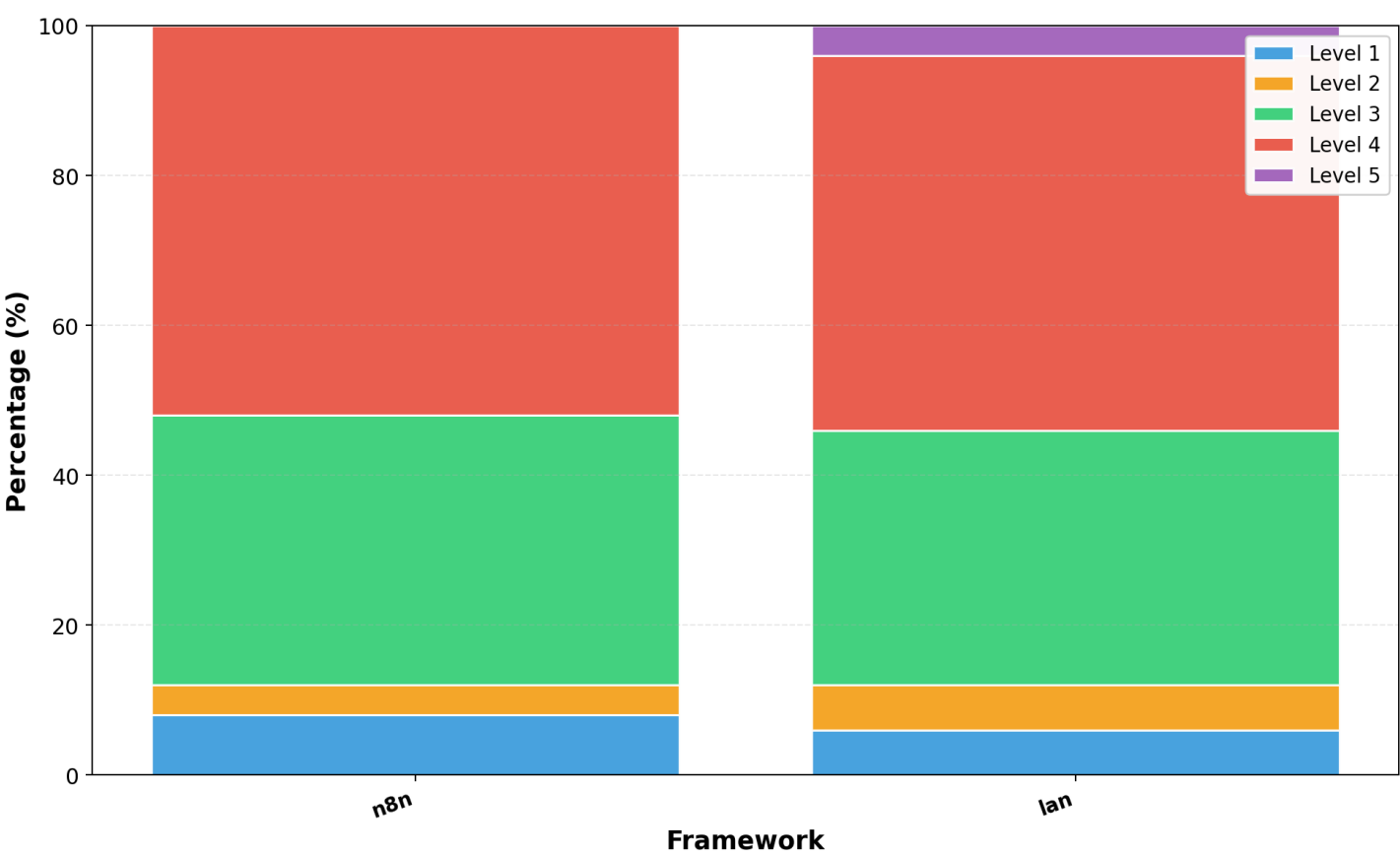
FULL COMPARATIVE ANALYSIS

Dimension	Metric	n8n	Ian	Difference / Note
Token Consumption	Prompt tokens (avg.)	6,452	866	-86.58%
Token Consumption	Completion tokens (avg.)	595	119	-79.90%
Token Consumption	Total tokens (avg.)	7,046	985	-86.02%
Token Consumption	Economic Efficiency (Correctness/USD)	3,692	22,121	-83.31%
Token Consumption	Token Efficiency (tok/pt)	2443.49	329.45	-86.52%
Latency	Execution time (ms)	29,510	8,309	-71.84%
Response Quality	RAGking Overall Score	49.4	62.9	-21.35%
Response Quality	Std. Dev. (RAGking Score)	8.51	11.8	-27.70%
Response Quality	Avg. correctness score	3.32	3.40	-2.35%
Response Quality	Cosine similarity (avg.)	0.63	0.64	-1.32%
Response Quality	Failure rate	6.00	0.00	-6.0
Response Quality	Level 1 responses	4	3	-1
Response Quality	Level 2 responses	2	3	-1
Response Quality	Level 3 responses	18	17	-1
Response Quality	Level 4 responses	26	25	-1
Response Quality	Level 5 responses	0	2	-2
Deployment	Deploy Time (hours)	3.00	6.00	-3.0
Deployment	Difficulty (1-5)	2.00	3.00	-1.0
Deployment	Maintenance (h/week)	8.00	1.00	-7.0

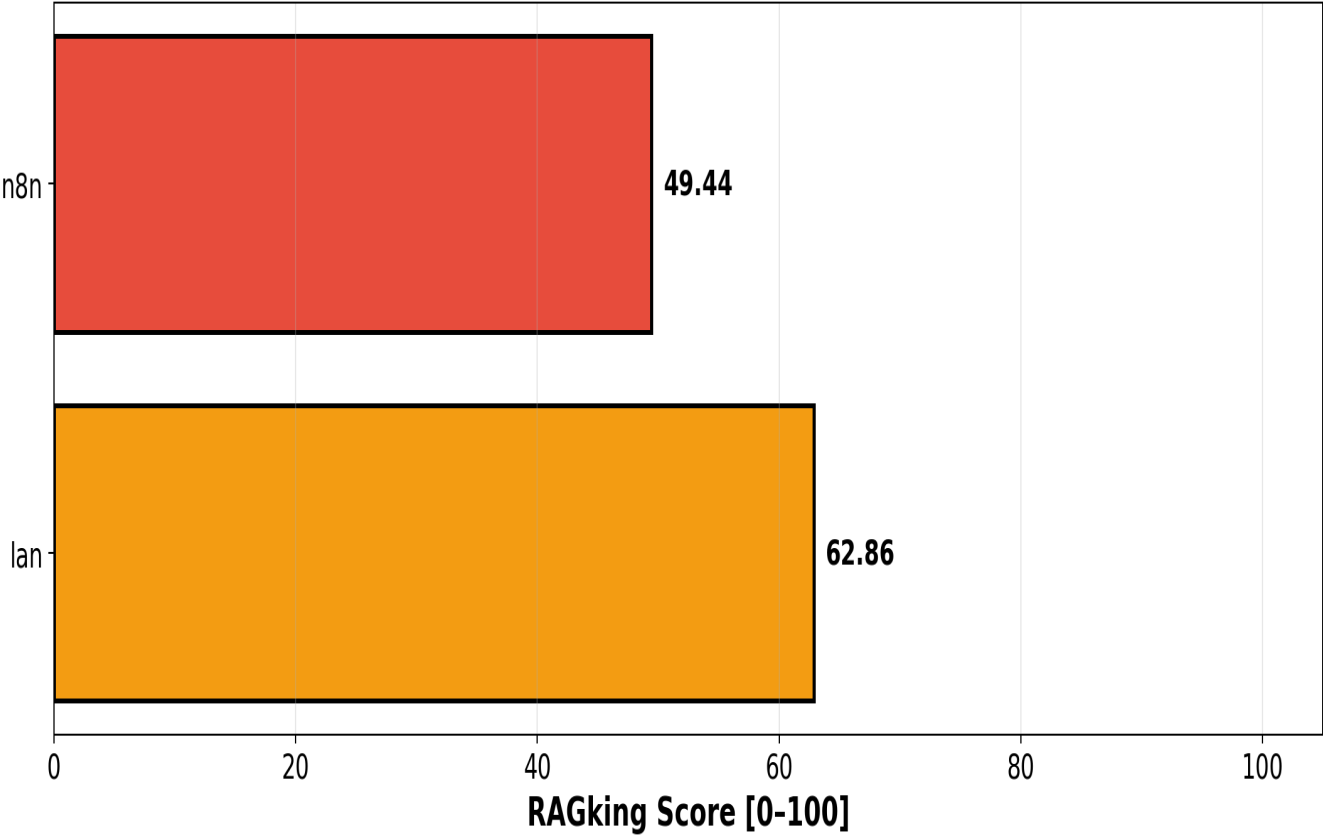
RESPONSE LEVEL DISTRIBUTION

Stacked percentages per framework to compare the share of Level 1–5 responses.

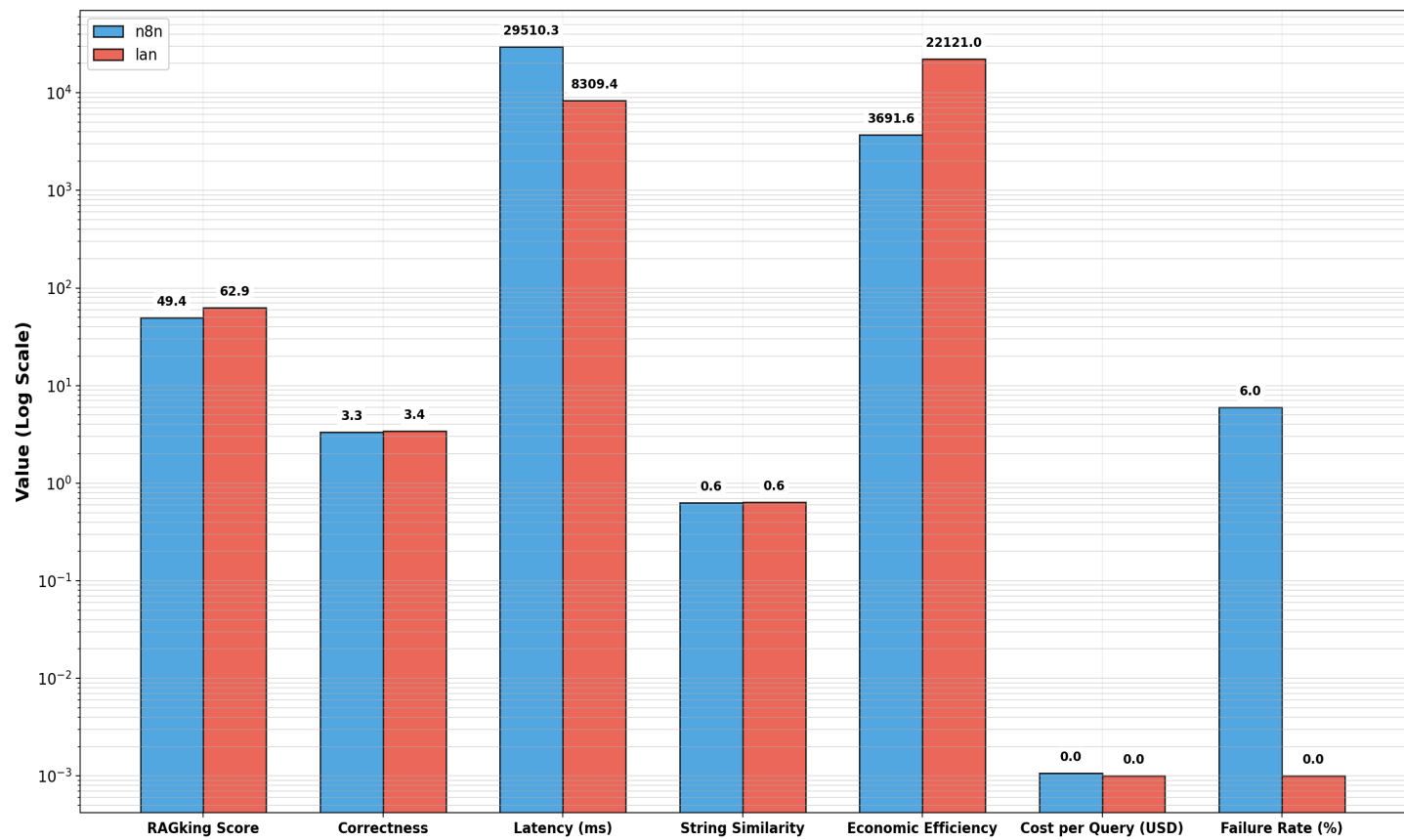
Distribution of Correctness Levels



OVERALL RAGKING SCORES

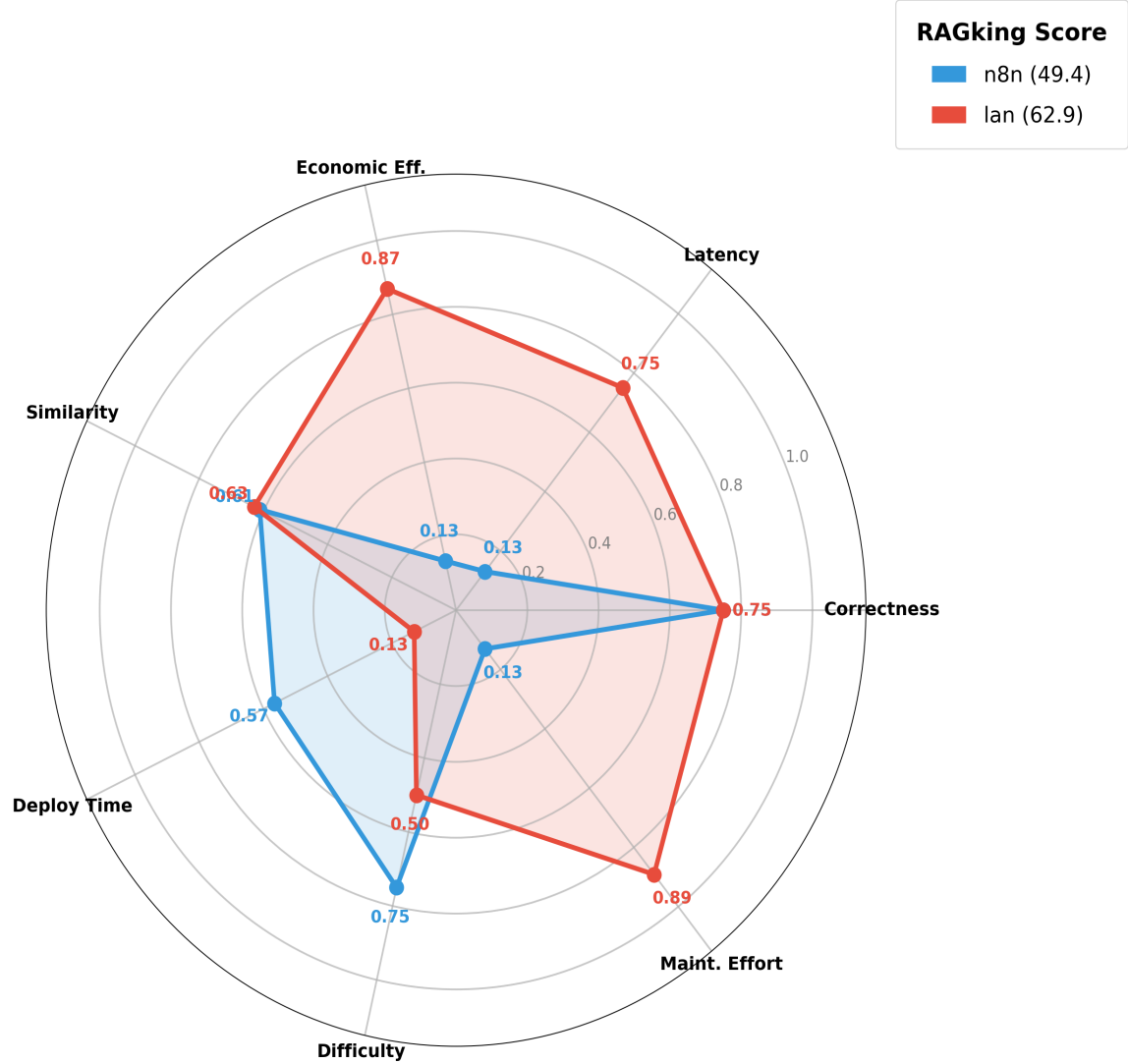


MULTI-METRIC COMPARISON



NORMALIZED METRICS PROFILE

Normalized [0–1]. Outer = better performance.



Values normalized to [0–1] · Best performer: outer ring

DISTRIBUTION COMPARISON

